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Inflation-Indexed Bonds and the Expectations Hypothesis

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TIPS, breakeven inflation, return predictability, bond risk premia

Abstract

This review empirically analyzes the expectations hypothesis (EH) in inflation-indexed (or real) bonds and in nominal bonds in the United States and in the United Kingdom. We strongly reject the EH in inflation-indexed bonds, and also confirm and update the existing evidence rejecting the EH in nominal bonds. This rejection implies that the risk premium on both real and nominal bonds varies predictably over time. We also find strong evidence that the spread between the nominal and the real bond risk premium, or the breakeven inflation risk premium, also varies over time. We argue that the time variation in real bond risk premia most likely reflects both a changing real interest rate risk premium and a changing liquidity risk premium, and that the variability in the nominal bond risk premia reflects a changing inflation risk premium. We estimate significant time series variability in the magnitude and sign of bond risk premia.

1. INTRODUCTION

This review conducts an empirical exploration of the magnitude and time variation of risk premia in inflation-indexed and nominal government bonds, using data on U.S. Treasury bonds and UK gilts. Understanding bond risk premia is fundamental in thinking about the term structure of interest rates. It is also of first-order importance for bond issuers, given that public debt constitutes one of the main sources of government financing, and for users, whether central banks or investors. Central banks use government bonds as a key instrument in the execution of monetary policy, whereas investors use them as the anchor of their fixed income allocations.

The most common form of government bonds are nominal bonds that pay fixed coupons and principal. However, in recent times governments around the world, including the U.S. Treasury, have started issuing significant amounts of inflation-indexed bonds (Campbell et al. 2009). Inflation-indexed bonds, which in the United States are known as Treasury Inflation-Protected Securities (TIPS), are bonds whose coupons and principal adjust automatically with the evolution of a consumer price index. [In the United States, TIPS payments are linked to the Consumer Price Index for All Urban Consumers (CPI-U). The relevant index in the United Kingdom is the Retail Price Index (RPI).] They aim to pay investors a fixed inflation-adjusted coupon and principal. For this reason they are also known as real bonds, and their yields are typically considered the best proxy for the term structure of real interest rates in the economy.

Although government bonds in large stable economies are generally free from default risk, they expose investors to other risks. Investors holding either inflation-indexed or nominal government bonds are exposed to the risk of changing real interest rates. For any investor the riskless asset is an inflation-indexed bond whose cash flows match his consumption plan (Campbell & Viceira 2001, Wachter 2003). If future real interest rates are uncertain, investors will view bonds not matching the timing and length of their consumption plans as risky, leading to a risk premium for holding such bonds. This real interest rate risk premium will be a function of investors' risk tolerance, and it can be time varying if investors' tolerance for risk changes over the business cycle (Campbell & Cochrane 1999, Wachter 2006). A time-varying correlation of real interest rates with investor well-being can also make the real interest rate risk premium vary over time (Campbell et al. 2010).

In addition to real interest rate risk, nominal government bonds expose investors to inflation risk. When future inflation is uncertain, the coupons and principal of nominal bonds can suffer from the eroding effects of inflationary surprises. If inflation is negatively correlated with economic conditions, as in times of stagflation, the real payoffs of nominal bonds will tend to decline when economic conditions worsen. Risk-averse investors will therefore demand a positive inflation risk premium for holding nominal bonds. But if inflation is positively correlated with economic conditions, nominal bonds will have hedging value to risk-averse investors (Piazzesi & Schneider 2007, Campbell et al. 2010). By contrast, inflation-indexed bonds are not exposed to inflation risk, given that their coupons and principal adjust automatically with inflation. [Tax regulations in some countries, including the United States, make the after-tax income and capital gains from inflation-indexed bonds not fully inflation indexed. This effect can be exacerbated at times of high accelerating inflation. See Section 2.]

The starting point of our empirical investigation of bond risk premia is the expectations hypothesis (EH) of interest rates. The EH postulates that the risk premium on long-term bonds, or the expected excess return on long-term bonds over short-term bonds, should be constant over time. If the EH holds for inflation-indexed bonds, this implies that the real interest rate risk premium is constant. In that case the yield on long-term inflation-indexed bonds is equal to the average expected short-term real interest rate over the life of the bond plus a constant. Investors cannot earn predictable returns by shifting between long-maturity and short-maturity real bonds. The implications of the EH for nominal bonds are stronger. If it holds, both the inflation risk premium and the real interest rate risk premium are constant (unless we are in the unlikely case where time variation in the inflation risk premium and the real interest rate risk premium exactly cancel each other out). In that case the yield on long-term nominal bonds is equal to the average expected future short-term nominal interest rate up to a constant. A rejection of the nominal expectations hypothesis can be the result of a time-varying inflation risk premium, a time-varying real interest rate risk premium, or both. Without independent observation of real bond prices it is hard to distinguish between those sources of time variation in nominal bond risk premia.

In our analysis we adopt a flexible empirical approach that does not rely on a tightly parameterized model. [See Evans (2003), Buraschi & Jiltsov (2005), Adrian & Wu (2009), Campbell et al. (2010), and Christensen et al. (2010) for formal models of the term structure of interest rates that analyze and estimate inflation and real interest rate risk premia using data on both real and nominal bonds.]

The EH has been tested and rejected on U.S. nominal Treasury bonds numerous times, but previous tests for inflation-indexed bonds only had significantly shorter samples at their disposal and were not able to reject the expectations hypothesis (Barr & Campbell 1997). Campbell & Shiller (1991) present regression results for different combinations of maturities and holding periods and resoundingly reject the expectations hypothesis for U.S. nominal bonds. Fama & Bliss (1987), Cochrane & Piazzesi (2005), and others have also presented robust empirical evidence that nominal Treasury bond risk premia vary over time. However, Campbell (1999) presents evidence that the expectations hypothesis is harder to reject on nominal government bonds in a cross section of other developed economies.

A number of recent papers have investigated the liquidity properties of inflation-indexed bonds in the U.S. Although nominal Treasury bonds are among the most liquid securities, empirical studies have found that TIPS tend to be significantly less liquid. [See for instance Shen (2006), D'Amico et al. (2008), Haubrich et al. (2011), Fleming & Krishnan (2009), Dudley et al. (2009), Fleckenstein et al. (2010), and Gürkaynak et al. (2010).] It therefore seems plausible to think that inflation-indexed bonds should not only reflect real long-term interest rates but might also be subject to liquidity differentials as compared to nominal bonds. These liquidity differentials might also lead to return predictability in inflation-indexed bonds and yield a rejection of the EH. Pflueger & Viceira (2011) decompose the return predictability in inflation-indexed bonds into a component attributable to real rate risk premia and one attributable to liquidity.

The structure of this review is as follows. Section 2 describes the mechanics of inflation-indexed bonds. Section 3 formalizes the EH of the term structure of interest rates and expected inflation. Section 4 tests the EH in real and nominal bonds. Section 5 provides evidence on real and nominal bond return predictability from the tent-shaped linear

combination of nominal interest rates proposed by Cochrane & Piazzesi (2005). Section 6 shows estimates of bond risk premia and their variation over time. Finally, Section 7 offers some concluding remarks and suggestions for future research.

2. INFLATION-INDEXED BONDS

Inflation-indexed bonds have been available in the United Kingdom since 1983 and in the United States since 1997. U.S. inflation-indexed bonds are called TIPS. They are designed to approximate real bonds with payouts that are constant despite inflation surprises. They are quoted in terms of a real interest rate and are issued mostly at long maturities greater than 10 years. The principal on these bonds grows with a prespecified price index, which in the United States is the CPI-U and in the United Kingdom is the RPI. The coupons are equal to the inflation-adjusted principal on the bond times a fixed coupon rate. Thus the coupons on these bonds also fluctuate with inflation.

Of course, the price index might not grow over time. For instance the CPI-U decreased by almost 4% between September 2008 and December 2008. In the United States, the final payment of principal on TIPS is protected against deflation, and it can never be smaller than the stated nominal value at issuance. However, TIPS coupons are not: The inflation-adjusted value of the principal for coupon computation purposes can fall below the initial value at issuance. In contrast, neither the principal nor the coupons on inflation-linked gilts in the United Kingdom are protected from deflation.

Further details complicate the pricing of these bonds. Because inflation figures in the United States and in the United Kingdom are published with a lag, the principal value of inflation-indexed bonds adjusts with a three-month lag. UK inflation-linked gilts that were issued prior to the financial year 2005–2006 follow an eight-month lagged indexing procedure, whereas more recent issues follow a three-month lagged methodology. The tax treatment of these bonds also differs. In the United Kingdom principal adjustments of inflation-linked gilts are not taxed. This gives inflation-linked gilts a tax advantage over nominal gilts, a larger share of whose cash flows come in the form of taxable nominal coupon payments. In the United States, however, inflation adjustments of principal are considered ordinary income for tax purposes. As a result the tax obligations associated with holding a TIPS increase when inflation is high so that on an after-tax basis U.S. TIPS are not fully indexed to inflation. More details on TIPS can be found in Viceira (2001), Roll (2004) and Gürkaynak et al. (2010). Campbell & Shiller (1996) offer a discussion of the taxation of inflation-indexed bonds. Campbell et al. (2009) provide an overview of the history of inflation-indexed bonds in the United States and the United Kingdom.

3. THE EXPECTATIONS HYPOTHESIS

The EH of the term structure of interest rates says that the excess return on an n -period bond over a one-period bond should be constant over time. There should not be any particularly good time to hold short-term or long-term bonds. Equivalently, the expectations hypothesis says that if short yields are anticipated to rise in the future then this should already be reflected in current long yields. The expectations hypothesis is usually stated for nominal bonds. We formulate expectations hypotheses for nominal bonds, real bonds, and for inflation expectations. We show that these different interpretations of the

expectations hypothesis are closely related to real interest rate risk, inflation risk, and liquidity premia and derive empirical predictions that we test subsequently.

3.1. Bond Notation and Definitions

We start by establishing some notation and definitions that are used throughout the review. We denote by $p_{n,t}^{\$}$ the log price of a zero-coupon n -period nominal bond, and by $y_{n,t}^{\$}$ the bond's log (or continuously compounded) yield. For zero-coupon bonds, log price and yield are related according to

$$y_{n,t}^{\$} = -\left(\frac{1}{n}\right)p_{n,t}^{\$}. \quad (1)$$

The yield spread is the difference between a long-term yield and a short-term yield, $s_{n,t}^{\$} = y_{n,t}^{\$} - y_{1,t}^{\$}$.

The log return on a zero-coupon n -period bond if it is held for one period and sold before maturity is given by the change in its log price, that is,

$$\begin{aligned} r_{n,t+1}^{\$} &= p_{n-1,t+1}^{\$} - p_{n,t}^{\$} \\ &= ny_{n,t}^{\$} - (n-1)y_{n-1,t+1}^{\$}, \end{aligned} \quad (2)$$

where the second equality follows immediately from Equation 1.

We use the superscript *TIPS* to denote the corresponding quantities for both U.S. and UK inflation-indexed bonds. Inflation-indexed bonds are commonly quoted in terms of real yields. That is, $p_{n,t}^{TIPS}$ is the real log price of an indexed bond, $y_{n,t}^{TIPS}$ is the real yield, and $r_{n,t+1}^{TIPS}$ is the real one-period log return. The nominal one-period log return on an inflation-indexed bond is then given by $r_{n,t+1}^{TIPS} + \pi_{1,t}$, where $\pi_{1,t}$ denotes the one-period log inflation rate from period t to period $t+1$.

3.2. Nominal Expectations Hypothesis

The nominal EH states that the expected log excess return on long-term nominal bonds over short-term nominal bonds, or bond risk premium, is constant over time:

$$E_t[r_{n,t+1}^{\$} - y_{1,t}^{\$}] = \mu_n^{\$}, \quad (3)$$

where the constant bond risk premium $\mu_n^{\$}$ can depend on maturity n .

We follow Campbell & Shiller (1991) in testing the log EH. Formulating the EH in terms of simple returns as opposed to log returns can be thought of as similar to investor risk neutrality.¹ Expected log returns are not the same as expected simple returns, the difference reflecting one half times the variance of log returns. Therefore, the EH in log returns is not equivalent to the EH in simple returns. However, the EH in simple returns over one holding period, such as one quarter, is also not consistent with the EH in simple returns over a different holding period, such as one year, again because of a term adjusting

¹Another version of the EH, the so-called pure expectations hypothesis (PEH), considers a formulation of Equation 3 in terms of simple returns as opposed to log returns, and sets expected excess simple returns to zero (Campbell et al. 1997). The intuition of the PEH is that if investors are risk neutral then they should adjust positions until the expected one-period returns for short nominal bonds and long nominal bonds are equalized. The log EH (Equation 3) is less constraining in that it allows for a nonzero bond risk premium.

for the variance of returns. We therefore prefer to formulate the EH in logs, because the log EH for one holding period is consistent with the log EH for any other holding period.

The EH can be represented in several different ways that can be obtained by simple algebraic manipulation of Equations 2 and 3. [For a detailed discussion of equivalent formulations of the EH, see Campbell et al. (1997, chapter 10) or Cochrane (2005, chapter 19).] A popular equivalent representation of the nominal EH relates the yield on an n -period zero-coupon nominal bond at time t to expected future short-term nominal interest rates:

$$y_{n,t}^{\$} = \theta_n^{\$} + \frac{1}{n} E_t \sum_{i=0}^{n-1} y_{1,t+i}^{\$}. \quad (4)$$

Equation 4 says that the current n -period yield should be equal to the expected average short yields over its maturity up to a time-invariant constant $\theta_n^{\$}$. The constant $\theta_n^{\$}$ is simply the average of bond risk premia for maturities up to n periods, that is, $\theta_n^{\$} = \sum_{i=2}^n \mu_i^{\$} / n$. A second equivalent representation of the nominal EH relates changes in long-term yields to the yield spread

$$E_t [y_{n-1,t+1}^{\$} - y_{n,t}^{\$}] = \left(\theta_{n-1}^{\$} - \frac{n}{n-1} \theta_n^{\$} \right) + \frac{1}{n-1} s_{n,t}^{\$}. \quad (5)$$

Although these alternative equivalent representations of the EH provide useful intuition to understand the meaning and implications of the EH, we choose to work with the return-based definition (Equation 3) in our empirical exploration of the EH. This approach allows for transparent interpretation of empirical results in terms of return predictability, and it is flexible enough to easily accommodate a complementary analysis of liquidity premia and supply pressures in the bond market.

The EH says that $r_{n,t+1}^{\$} - y_{1,t}^{\$}$ cannot be predicted. However, early tests of the EH based on Equation 5 found robust evidence that the nominal term spread—or an equivalent combination of forward rates—predicts nominal excess returns positively (Fama & Bliss 1987, Campbell & Shiller 1991). That is, whenever the term spread is high the risk premium on long nominal bonds is higher. [This is equivalent to finding a negative slope in a regression of changes in the yield on long-term bonds on $s_{n,t}^{\$} / (n-1)$.] Building on this prior work, we test in our data whether the term spread contains a time-varying risk premium by running the regression test

$$r_{n,t+1}^{\$} - y_{1,t}^{\$} = \alpha^{\$} + \beta^{\$} s_{n,t}^{\$} + \varepsilon_{t+1}^{\$}, \quad (6)$$

where $\beta^{\$} = 0$ under the null that the EH holds. Of course, failing to reject $\beta^{\$} = 0$ in Equation 6 does not imply that we fail to reject the EH, as other state variables might forecast bond excess returns. Thus we also include in Equation 6 other variables that have been shown to forecast bond excess returns in our empirical analysis.

3.3. Real Expectations Hypothesis

The EH has traditionally been formulated and tested in terms of nominal bonds but it appears at least as plausible to formulate the hypothesis in terms of real bonds. The nominal EH supposes that the bond risk premium on nominal bonds, consisting of both

inflation risk and real interest rate risk, is constant over time. The EH for inflation-indexed bonds is strictly weaker in that it only supposes that real interest rate risk is constant.

Expressed in terms of returns, the EH for inflation-indexed zero-coupon bonds says that

$$E_t[r_{n,t+1}^{TIPS} - y_{1,t}^{TIPS}] = \mu_n^{TIPS}. \quad (7)$$

Analogously to the nominal EH we test the real EH by testing whether the real term spread predicts excess returns on real bonds:

$$r_{n,t+1}^{TIPS} - y_{1,t}^{TIPS} = \alpha^{TIPS} + \beta^{TIPS} s_{n,t}^{TIPS} + \varepsilon_{t+1}^{TIPS}, \quad (8)$$

where $s_{n,t}^{TIPS} = y_{n,t}^{TIPS} - y_{1,t}^{TIPS}$ is the TIPS bond spread. The real EH hypothesis implies that the coefficient of real excess log returns of inflation-indexed bonds on the real term spread should be zero. If $\beta^{TIPS} \neq 0$ then we can infer that the real yield reflects time-varying real risk premia and μ_n^{TIPS} is time varying. The TIPS bond spread is a natural candidate regressor due to its similarity to the nominal bond spread. Because TIPS are not exposed to inflation surprises, the TIPS yield spread should not reflect inflation risk, although it might reflect other risk premia such as real interest rate risk and liquidity premia. Hence, if any of these premia are important in driving the rejection of the nominal EH they would be likely to be reflected in terms of a nonzero coefficient β^{TIPS} .

3.4. Breakeven Inflation and the Inflation Expectation Hypothesis

We now look at the difference between nominal and indexed yields, known by bond market participants as breakeven inflation:

$$b_{n,t} = y_{n,t}^{\$} - y_{n,t}^{TIPS}. \quad (9)$$

Most simply, n -period breakeven inflation is the inflation rate over the next n periods that would make a nominal bond and an indexed bond of maturity n earn the exact same buy-and-hold return. The nominal bond outperforms the inflation-indexed bond if realized inflation over the life of the bonds turns out to be smaller than breakeven inflation, and underperforms it if realized inflation turns out to be larger.

Bond market participants often use breakeven inflation as a measure of expected inflation. However, the identification of breakeven inflation with expected inflation is not entirely correct. To understand this point, it is useful to think about the components of bond yields, both nominal and inflation indexed. Economic logic, formally corroborated by models of the term structure of interest rates,² suggests that we can decompose the yield—or equivalently, the price—on an inflation-indexed bond into a component that reflects current expectations of future real interest rates, and a component that reflects a real interest rate risk premium. Similarly, we can think of the yield on a nominal bond as composed of the yield on a real bond plus additional components reflecting expectations of future inflation and an inflation risk premium. Thus the spread between both yields, or breakeven inflation, reflects both expected inflation and the inflation risk premium embedded in the nominal bond yield.

²Again, see Adrian & Wu (2009), Buraschi & Jiltsov (2004), Campbell et al. (2010), Christensen et al. (2010) and Evans (2003) for formal models of the term structure of interest rates that analyze and estimate inflation and real interest rate risk premia using data on both real and nominal bonds.

If institutional, behavioral, or liquidity factors affect the nominal bond market and the inflation-indexed bond market differently, breakeven inflation will also reflect the differential impact of these factors on yields (Pflueger & Viceira 2011). For example, we think of the market for inflation-indexed bonds to be less liquid than the market for nominal bonds. If investors apply a liquidity discount to the price of inflation-indexed bonds, or a liquidity premium to the price of nominal bonds, breakeven inflation will be lower than it would be otherwise, given that prices and yields move inversely.³ When changes in the liquidity differential are correlated with aggregate economic conditions, breakeven inflation will also reflect an additional liquidity risk premium.

Of course, expected inflation, the inflation risk premium, the liquidity differential, and the liquidity risk premium need not be constant over time, causing realized breakeven inflation to move over time. More importantly, time variation in the inflation risk premium or in the liquidity premium can also cause the expected excess return on breakeven inflation, or the difference between the excess return on nominal bonds and the excess return on inflation-indexed bonds of identical maturity, to vary over time. Mechanically, the excess return on breakeven inflation is given by

$$r_{t+1}^b \equiv (r_{n,t+1}^{\$} - y_{1,t}^{\$}) - (r_{n,t+1}^{TIPS} - y_{1,t}^{TIPS}) = nb_{n,t} - (n-1)b_{n-1,t+1} - b_{1,t}. \quad (10)$$

Under the assumption of constant inflation and liquidity risk premia, the left-hand side of Equation 10 equals a constant plus the expression $n\pi_{n,t}^e - (n-1)\pi_{n-1,t+1}^e - \pi_{1,t}$, where $\pi_{n,t}^e$ denotes n -period expected inflation at time t . This expression is zero when inflation expectations are rational. To see this, note that realized inflation verifies

$$n\pi_{n,t} - (n-1)\pi_{n-1,t+1} - \pi_{1,t} = 0, \quad (11)$$

given that both $n\pi_{n,t}$ and $(n-1)\pi_{n-1,t+1} + \pi_{1,t}$ denote cumulative inflation from time t to time $t+n$. Therefore under rational expectations, Equation 11 must also hold ex ante.

We call the joint hypothesis of rational inflation expectations and a constant inflation risk premium the inflation expectations hypothesis. By analogy with our tests of the nominal and real EH, we run the regression

$$r_{t+1}^b = \alpha^b + \beta^b s_{n,t}^b + \varepsilon_{t+1}^b, \quad (12)$$

where $s_{n,t}^b = b_{n,t} - b_{1,t}$ is the breakeven inflation spread, and test whether $\beta^b = 0$. If the inflation risk premium or the liquidity risk premium is time varying, and they are correlated with the breakeven inflation spread, we would expect to find a non-zero regression slope coefficient β^b in Equation 12. In particular, the breakeven spread $s_{n,t}^b$ should reflect the inflation risk premium contained in the nominal yield spread $s_{n,t}^{\$}$.

Because the breakeven inflation spread, the nominal term spread, and the real term spread are mechanically related by $s_{n,t}^{\$} = s_{n,t}^b + s_{n,t}^{TIPS}$, it also makes sense to test whether both the real term spread and the breakeven inflation spread jointly forecast the return

³Campbell et al. (2009) document an episode of "flight to liquidity" during the recent financial crisis. In the fall of 2008 the price of nominal Treasury bonds increased rapidly, while the price of TIPS declined, causing a dramatic narrowing of breakeven inflation, which at some point became even negative. They provide evidence that this change in prices did not reflect a sudden change in the outlook for inflation toward massive deflation, but rather an increase in the liquidity differential between both markets, as investors around the world flew into nominal Treasuries.

on breakeven inflation. It is important to note that neither Equation 12 nor the expanded version of the equation that includes $s_{n,t}^{TIPS}$ is redundant with respect to the standard EH regressions (Equations 8 and 6), except of course in the trivial case where the spreads do not forecast bond excess returns and thus all slope coefficients are zero.

4. TESTING THE EXPECTATIONS HYPOTHESIS IN REAL AND NOMINAL GOVERNMENT BONDS

4.1. Data

We conduct tests of the real and nominal EH using government bond data from both the United States and United Kingdom. For the United States we use an expanded version of the Gürkaynak, Sack, and Wright (GSW henceforth) data set (Gürkaynak et al. 2007, 2010). GSW have constructed a zero-coupon yield curve starting in January 1961 for nominal bonds and starting in January 1999 for TIPS by fitting a smoothed yield curve. We expand their data back to 1951 using the McCulloch & Kwon (1993) data for U.S. nominal zero-coupon yields from January 1951 through December 1960. The GSW data set contains constant maturity yields for maturities of 2 to 20 years. Our empirical tests focus on the 10-year nominal and real yields, because this maturity bracket has the longest and most continuous history of TIPS outstanding. We measure U.S. inflation with the all-urban seasonally adjusted CPI, and the short-term nominal interest rate with the three-month T-bill rate from the Fama-Bliss riskless interest rate file from the Center for Research in Security Prices. TIPS payouts are linked to the all-urban non-seasonally adjusted CPI, and our results are slightly stronger when using the non-seasonally adjusted CPI instead.

For the United Kingdom we use zero-coupon yield curves from the Bank of England. Anderson & Sleath (2001) describe the spline-based techniques used to estimate the yield curves. Nominal yields are available starting in 1970 for 0.5 to 20 years to maturity. Real yields are available starting in 1985 for 2.5 to 25 years to maturity. We focus on the 20-year nominal and real yields because they are available from 1985, whereas other maturities are available only since 1991. We measure inflation by the non-seasonally adjusted RPI, which serves as the measure of inflation for inflation-indexed bond payouts. We obtain the three-month UK nominal interest rate from Datastream (UKBTND).

In all regressions we approximate $y_{n-1,t+1}^{\$}$ and $y_{n-1,t+1}^{TIPS}$ with $y_{n,t+1}^{\$}$ and $y_{n,t+1}^{TIPS}$. Because neither the United States nor the United Kingdom governments issue inflation-indexed bills, we need to resort to an empirical procedure to build a hypothetical short-term real interest rate. We describe this procedure in Section 4.2. Finally, although our yield data sets are available at a monthly frequency, we use quarterly overlapping returns to reduce the influence of high-frequency noise in observed inflation and short-term nominal interest rate volatility in our tests.

4.2. Construction of the Short-Term Real Interest Rate

Although three-month nominal T-bills are issued in the United States and in the United Kingdom, there exists no equivalent short-term instrument with fixed real payoffs. Apart

from technical difficulties, the demand would probably not exist simply because inflation risk in both countries has been historically negligible over such a short horizon. However, we need a proxy for a short-term real rate for our tests of the EH. We follow Campbell & Shiller's (1996) analysis of hypothetical TIPS to construct an ex ante measure of the short-term real interest rate.

We start by assuming zero inflation risk and liquidity premium over one quarter. Therefore, the ex ante short-term real interest rate is given by

$$y_{1,t}^{TIPS} = y_{1,t}^{\$} - \pi_{1,t}^e.$$

Next we assume that inflation expectations over the next quarter are rational and proxy for the ex ante real short rate as the fitted value from the regression of this quarter's realized real rate $y_{1,t}^{\$} - \pi_{1,t}$ onto last quarter's realized real rate $y_{1,t-1}^{\$} - \pi_{1,t-1}$, the nominal short rate $y_{1,t}^{\$}$, and annual inflation up to time t .

Table 1 shows the monthly predictive regressions for the United States and the United Kingdom. It reports the point estimates of the slope coefficients as well as Newey-West heteroskedasticity and autocorrelation consistent standard errors with four lags in parentheses. The table shows that the main determinant of the ex ante real rate is the nominal rate, with a positive coefficient of approximately 0.5 in both the United States and the United Kingdom. The regressions can explain 44% of the real interest rate variation in the United States and 18% of the real interest rate variation in the United Kingdom, and the regressors are jointly significant in both regressions.

Figure 1 (see color insert) shows the predicted and realized U.S. real short rates together with the nominal short rate. It shows that the predicted real short rate very much just follows the nominal short rate and smoothes out fluctuations in the ex post real rate caused by short-term volatility in realized inflation. The estimate is conservative in the sense that it barely relies on lagged realized inflation and it does not attempt to predict high-frequency fluctuations in inflation.

Table 1 Forecasted real short rate ^a		
$y_{1,t}^{\$} - \pi_t$	U.S.	UK
$y_{1,t}^{\$}$	0.57** (0.22)	0.46 (0.29)
$y_{1,t-1}^{\$} - \pi_{t-1}$	0.08 (0.08)	-0.11 (0.07)
$(\pi_{t-4} + \pi_{t-3} + \pi_{t-2} + \pi_{t-1})/4$	0.08 (0.09)	0.03 (0.09)
<i>p</i> -value	0.00	0.00
<i>R</i> ²	0.44	0.18

^aOverlapping quarterly real short rate returns onto the nominal short rate and last quarter's real short rate return and inflation over the past year.
Monthly data 1982.1–2009.12. π_t denotes inflation from time t to time $t + 1$.
Newey-West standard errors with four lags in parentheses.
* and ** denote significance at the 5% and 1% levels, respectively.
p-value of the F-test for no predictability.

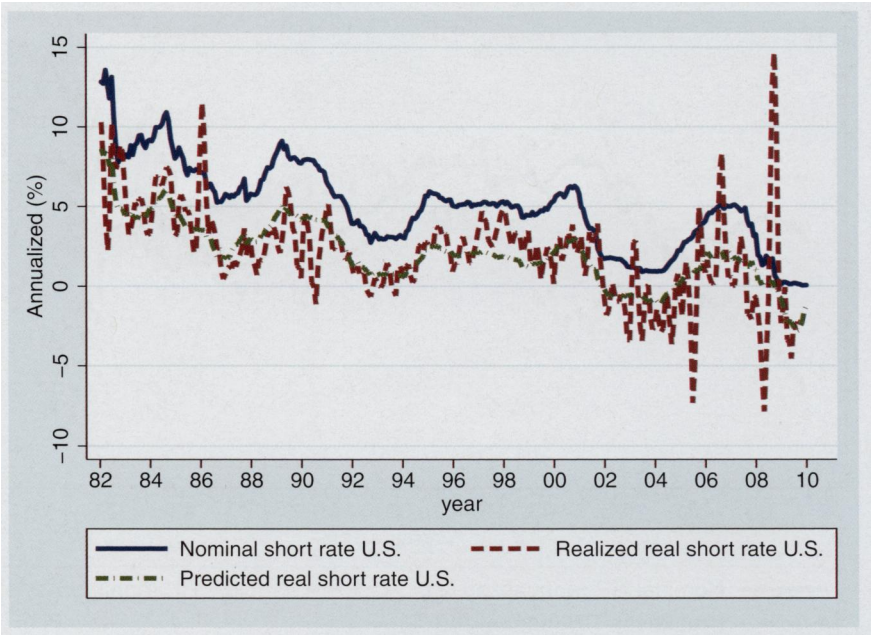


Figure 1
U.S. realized and predicted real short rate.

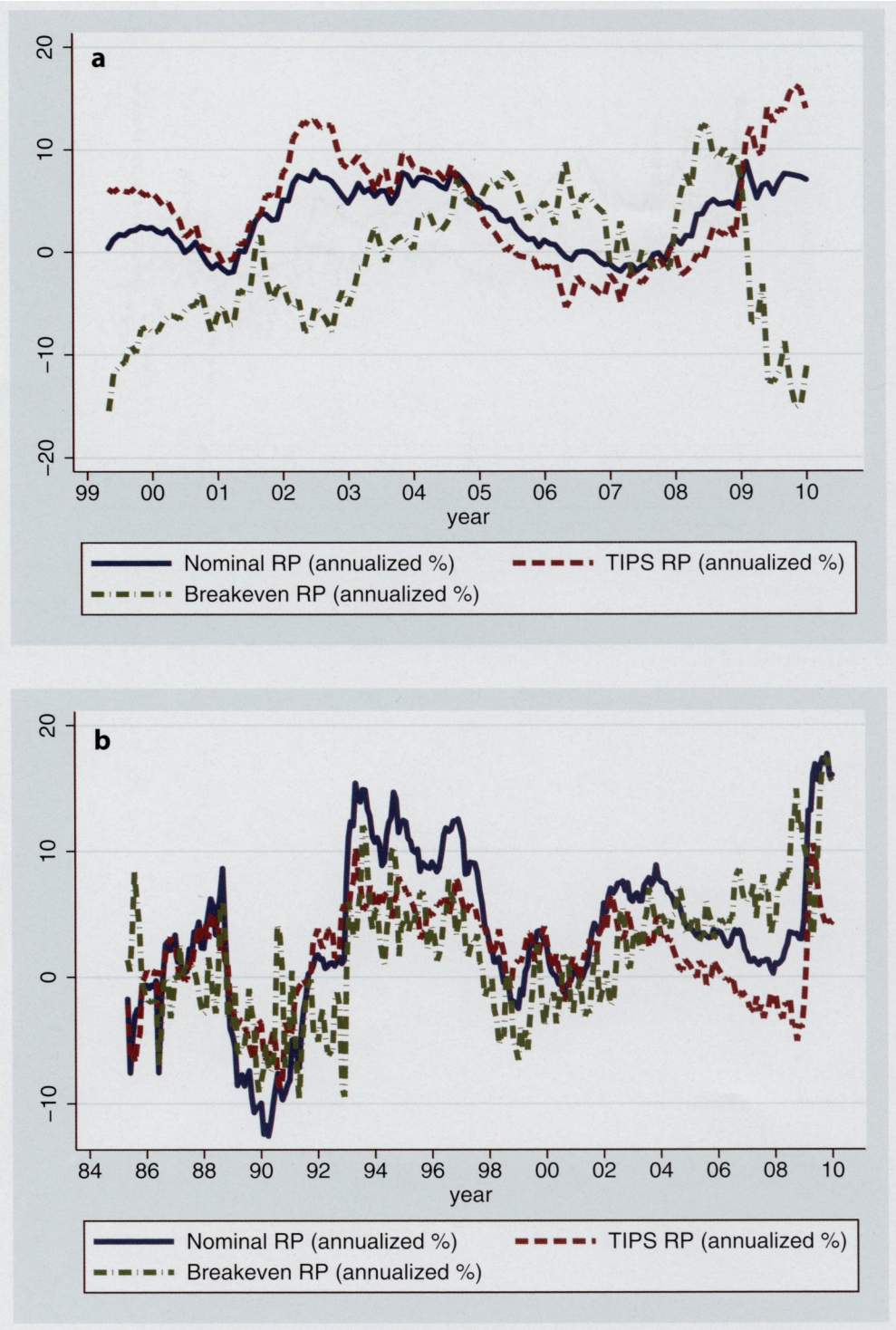


Figure 2
a) Fitted U.S. risk premia. *b)* Fitted UK risk premia. Abbreviations: RP, risk premia; TIPS, Treasury inflation-protected securities.

Table 2 Sample moments of inflation, interest rates, and bond returns^a

A: U.S. 10 YR									
	π_t	$y_{1,t}^{\$}$	$y_{1,t}^{TIPS}$	$y_{n,t}^{\$} - y_{1,t}^{\$}$	$y_{n,t}^{TIPS} - y_{1,t}^{TIPS}$	$b_{n,t} - b_{1,t}$	$xr_{n,t+1}^{\$}$	$xr_{n,t+1}^{TIPS}$	$r_{n,t+1}^b$
Mean	2.56	2.87	0.50	1.86	2.04	-0.17	3.26	4.16	-0.91
Std	1.50	0.96	0.72	0.72	0.61	0.37	8.57	7.67	7.22
Return correlations									
	$xr_{n,t+1}^{\$}$	$xr_{n,t+1}^{TIPS}$	$r_{n,t+1}^b$						
$xr_{n,t+1}^{\$}$	1	0.61	0.54						
$xr_{n,t+1}^{TIPS}$	.	1	-0.34						
$r_{n,t+1}^b$	.	.	1						
B: UK 20 YR									
	π_t	$y_{1,t}^{\$}$	$y_{1,t}^{TIPS}$	$y_{n,t}^{\$} - y_{1,t}^{\$}$	$y_{n,t}^{TIPS} - y_{1,t}^{TIPS}$	$b_{n,t} - b_{1,t}$	$xr_{n,t+1}^{\$}$	$xr_{n,t+1}^{TIPS}$	$r_{n,t+1}^b$
Mean	3.47	6.55	3.14	-0.24	-0.37	0.13	3.47	1.66	1.81
Std	1.84	1.52	0.77	0.97	0.58	0.57	14.67	9.25	11.97
Return correlations									
	$xr_{n,t+1}^{\$}$	$xr_{n,t+1}^{TIPS}$	$r_{n,t+1}^b$						
$xr_{n,t+1}^{\$}$	1	0.58	0.78						
$xr_{n,t+1}^{TIPS}$	.	1	-0.06						
$r_{n,t+1}^b$	.	.	1						

^aAnnualized (%). Monthly data of quarterly overlapping returns and inflation.
U.S. data are monthly 1999.4–2009.12. UK data are monthly 1985.4–2009.12.

$$\begin{aligned}xr_{n,t+1}^{\$} &= r_{n,t+1}^{\$} - y_{1,t}^{\$}\\xr_{n,t+1}^{TIPS} &= r_{n,t+1}^{TIPS} - y_{1,t}^{TIPS}\\r_{n,t+1}^b &= xr_{n,t+1}^{\$} - xr_{n,t+1}^{TIPS}\end{aligned}$$

Table 2 presents summary statistics for inflation, short-term nominal and real interest rates, nominal and real yield spreads, breakeven inflation, and bond returns for the United States (panel A) and the United Kingdom (panel B). Because the sample period and bond maturity in each table are different, it is hard to do comparisons across panels. Nonetheless, the average excess return on nominal bonds is similar across both countries, whereas the average excess return on inflation-indexed bonds has been larger in the United States. Bond return volatilities and correlations are generally comparable across both countries, controlling for maturity differentials. The average excess returns and volatilities reported in Table 2 imply sample Sharpe ratios on U.S. real and nominal bonds of 0.392 and 0.542, respectively. These are larger than the corresponding Sharpe ratios for UK real and nominal bonds, at 0.236 and 0.179, respectively.

4.3. The Nominal Expectations Hypothesis in the United States

Tables 3 reports tests of the nominal EH in the United States using our preferred return-based regression test (Equation 6). Thus we test whether nominal log excess

Table 3 Tests of the nominal expectations hypothesis: long sample^a

$xr_{n,t+1}^s$	1952.1–2009.12	1952.1–1987.2	1987.3–2009.12	1952.1–2009.12
$(y_{n,t}^s - y_{1,t}^s)$	3.57** (1.12)	4.85* (1.95)	1.81 (1.18)	4.85* (1.95)
$(y_{n,t}^s - y_{1,t}^s) \times d_{1987-2009}$				-3.04 (2.28)
$d_{1987-2009}$				0.01 (0.01)
p -value	0.00	0.01	0.13	0.01
R^2	0.05	0.07	0.02	0.06

^aOverlapping quarterly returns $r_{n,t+1}^s$ onto $y_{n,t}^s - y_{1,t}^s$. Monthly data 1952.1–2009.12.
 $d_{1987-2009}$ equals 1 in 1987.3–2009.12 and zero otherwise.
Newey-West standard errors with three lags in parentheses.
* and ** denote significance at the 5% and 1% levels, respectively.
 p -value of the F-test for no predictability.
 $xr_{n,t+1}^s$ is computed as $xr_{n,t+1}^s = r_{n,t+1}^s - y_{1,t}^s$.

returns are predictable from the nominal term spread. The objective of this table is to analyze changes in the predictability of nominal bond returns since Campbell & Shiller (1991) reported tests of the nominal EH. Campbell & Shiller found that they were able to clearly reject the EH for almost all of their maturity combinations for the sample period 1952–1987. Table 3 reruns those same regressions for our full sample period (1952.1–2009.12) with the 10-year constant maturity zero-coupon bond. For comparison, we also report them for the Campbell-Shiller sample period and the sample period from 1987 until present. [Campbell & Shiller (1991) used the McCulloch & Kwon (1993) data of zero-coupon yields for their entire period so that our results differ slightly from theirs due to our different data source.] The table reports the point estimates of the slope coefficients and Newey-West standard errors with 3 lags.

Table 3 shows that the full time period 1952–2009 yields an even stronger rejection of the EH than the earlier sample period 1952–1987. At the same time, if using the second part of the sample only, it is harder to reject the EH at conventional significance levels. Stock & Watson (2002) document a break in the mid-1980s in several macroeconomic time series. If the predictability of bond returns is linked to macroeconomic processes, it is conceivable that bond return predictability also experienced a break at the same time.

Following this intuition, the last column of Table 3 examines more closely whether there was a structural change in bond return predictability in 1987. We add the term spread interacted with a dummy for the second subperiod, $s_{n,t}^s \times d_{1987-2008}$, to the regression to allow for different slope coefficients before 1987 and after 1987. The interaction term does not enter significantly into the regression, indicating that we cannot reject the hypothesis of a stable relationship across subsamples. Moreover, the full sample period and the Campbell-Shiller sample period yield very similar regression coefficients and the coefficient is more accurately measured using the full sample period.

Thus the addition of the 1987–2009 period to the early sample period reinforces the empirical evidence toward rejection of the nominal EH. It also emphasizes the difficulty of detecting this type of bond return predictability in smaller sample sizes, even if the sample comprises more than 20 years of data. This qualification is particularly important for our subsequent analysis of the real EH, because even our longest series of inflation-indexed bonds only spans a 24-year period from 1985 to 2009.

4.4. Expectations Hypothesis, Real and Nominal

Table 4 presents our main regression tests for the nominal, real, and inflation EH in the United States and in the United Kingdom. Columns 2 through 5 in each panel report coefficients from the return-based regressions, Equations 6, 8, and 12, and the expanded

Table 4 Tests of the real and nominal expectations hypothesis: common sample^a

A: U.S. 10 YR				
	$xr_{n,t+1}^S$	$xr_{n,t+1}^{TIPS}$	$r_{n,t+1}^b$	$r_{n,t+1}^b$
$y_{n,t}^S - y_{1,t}^S$	2.17 (1.49)			
$y_{n,t}^{TIPS} - y_{1,t}^{TIPS}$		4.45** (1.37)		−3.10* (1.22)
$b_{n,t} - b_{1,t}$			7.24* (3.13)	7.43* (2.45)
<i>p</i> -value	0.15	0.00	0.02	0.01
<i>R</i> ²	0.03	0.12	0.13	0.20
B: UK 20 YR				
	$xr_{n,t+1}^S$	$xr_{n,t+1}^{TIPS}$	$r_{n,t+1}^b$	$r_{n,t+1}^b$
$y_{n,t}^S - y_{1,t}^S$	3.21* (1.53)			
$y_{n,t}^{TIPS} - y_{1,t}^{TIPS}$		3.16* (1.30)		−1.31 (2.05)
$b_{n,t} - b_{1,t}$			4.31* (1.99)	4.89* (2.33)
<i>p</i> -value	0.04	0.02	0.03	0.10
<i>R</i> ²	0.05	0.04	0.04	0.05

^aOverlapping quarterly returns. U.S. data are monthly 1999.4–2009.12. UK data are monthly 1985.4–2009.12.

Newey-West standard errors with three lags in parentheses.

* and ** denote significance at the 5% and 1% levels, respectively.

p-value of the F-test for no predictability.

$$\begin{aligned}
 xr_{n,t+1}^S &= r_{n,t+1}^S - y_{1,t}^S \\
 xr_{n,t+1}^{TIPS} &= r_{n,t+1}^{TIPS} - y_{1,t}^{TIPS} \\
 r_{n,t+1}^b &= xr_{n,t+1}^S - xr_{n,t+1}^{TIPS}.
 \end{aligned}$$

version of Equation 12, respectively. The data consist of monthly data of overlapping quarterly returns. Newey-West standard errors are based on three lags to adjust for overlapping returns.

Panel A reports the regression tests for the U.S. data from 1999 to 2009. According to the nominal EH, the coefficient in column 2 should be zero. We cannot reject the nominal EH over this particular time period at conventional significance levels. However, this rejection is somewhat marginal—the significance level is 15%—and the results in Table 3 indicate that this may well be related to the short sample size rather than a change in the predictive relationship. Column 3 of the panel tests whether excess returns on inflation-indexed bonds are predictable. The coefficient on the real spread is large compared to the coefficient on the nominal spread reported in column 2, and statistically significant at the 1% level. This strong rejection of the real EH is particularly striking in light of the weak rejection of the nominal EH in column 2. It suggests that factors specific to the TIPS market such as liquidity might be driving expected excess return on TIPS, besides real interest rate risk. (Results omitted here to save space show that, interestingly, the nominal term spread does not forecast TIPS excess returns, whereas the real term spread does not forecast nominal bond excess returns.)

Column 4 of panel A tests the inflation EH in the United States. We find that the breakeven spread predicts the difference in nominal and inflation-indexed bond excess returns. Assuming that bond market participants' inflation expectations are rational and that liquidity premia are constant, this result is consistent with a time-varying inflation risk premium. Column 5 also controls for the real term spread in the regression and shows that adding this variable does not affect the predictive power or the coefficient estimate of the breakeven spread. These results suggest that when the breakeven spread increases, inflation risk also increases and investors demand a higher inflation risk premium from nominal bonds.

Interestingly, the real term spread appears to predict breakeven returns negatively in the United States. Thus a widening of the real term spread forecasts a decrease in the spread between nominal bond risk premia and inflation-indexed bond risk premia. One might expect that if the real term spread proxies for the real interest rate risk premium, its coefficient should be zero; that is, increases in the real interest rate risk premium should affect approximately in the same proportion the prices of both inflation-indexed bonds and nominal bonds. The rejection of the null hypothesis that it is zero suggests that the effect of the real term spread on breakeven inflation returns might be related to liquidity factors (Pflueger & Viceira 2011).

Panel B in Table 4 reports the corresponding results for the UK bond market. The pattern of results is remarkably consistent with the results shown in panel A for the U.S. bond market, with the exception that the real term spread does not appear to predict breakeven returns.

Overall, the results in Table 4 provide strong support for the predictability of nominal and real bond returns and for the predictability of their difference in U.S. and UK bond data. These results provide support for the hypothesis that the risk premium on nominal bonds is driven by both a time-varying inflation risk premium and a time-varying real interest rate risk premium. An increase in breakeven inflation forecasts positively an increase in nominal bond risk premia relative to inflation-indexed bond risk premia. The U.S. results also show the striking result that the real term spread forecasts negatively the spread between the nominal bond risk premium and the inflation-indexed bond risk

premium. Pflueger & Viceira (2011) present empirical evidence that this is the result of a time-varying liquidity risk premium in inflation-indexed bonds.

5. COCHRANE AND PIAZZESI BOND RETURN PREDICTABILITY AND THE INFLATION RISK PREMIUM

In recent work Cochrane & Piazzesi (CP) show that a tent-shaped linear combination of nominal forward rates is a good predictor of excess nominal bond returns for a wide range of bond maturities (Cochrane & Piazzesi 2005). Their findings imply that nominal bond risk premia are high when intermediate-term nominal interest rates are high relative to both shorter-term and longer-term rates. In the context of a nonlinear model of the term structure of interest rates, Campbell et al. (2010) interpret their findings as reflecting a time-varying transitory inflation risk premium.

We explore this interpretation by examining whether the CP tent-shaped combination of nominal forward rates forecasts inflation-indexed bond excess returns in addition to nominal bond excess returns. If the CP factor reflects inflation risk premia, it should not predict returns on inflation-indexed bonds.

We construct the CP factor from one- to five-year Fama-Bliss zero-coupon nominal bond yields, available from the Center for Research in Security Prices. Let $f_{n,t}^{\$}$ denote the log one-year nominal forward rate at time t for loans between $n - 1$ and n years in the future. We obtain the CP factor using the optimal weights found in Cochrane & Piazzesi (2005) as $CP_t = -2.14 f_{1,t}^{\$} + 0.81 f_{2,t}^{\$} + 3.00 f_{3,t}^{\$} + 0.80 f_{4,t}^{\$} - 2.08 f_{5,t}^{\$}$. Unfortunately, we do not have enough richness in the term structure of TIPS rates to construct a CP variable based on TIPS yields. We also limit our analysis to the United States.

Panel A in Table 5 reproduces the CP predictability results for our data set, using the 1952–2009 sample period and a one-quarter forecasting horizon. Column 2 in the panel shows that CP is significant and forecasts nominal bond excess returns with a respectable R^2 of 4% at a quarterly horizon. However, column 3 in the panel shows that the variable loses its statistical significance once we control for the yield spread.

Panel B in Table 5 shows that CP does not forecast nominal bond excess returns, TIPS excess returns, and breakeven inflation over our 1999–2009 sample period. Panel B also shows that the inclusion of CP in the nominal and real EH regressions does not change our basic results. The factor enters significantly and with a positive sign only in the last column. Comparing this to column 5 of Table 4 shows that CP also increases the R^2 from 20% to 27%. When CP is high, nominal bond excess returns are expected to be larger than inflation-indexed bond excess returns. This result is consistent with Campbell et al.'s (2010) interpretation of CP as a proxy for a time-varying inflation risk premium.

6. HISTORICAL FITTED RISK PREMIA

We now look at the fitted risk premia to better understand the economic significance of the bond return predictability examined in this review. Table 6 shows the mean and standard deviation of the fitted excess log returns from the EH regressions shown in Table 4. We refer to average fitted excess log returns as risk premia, omitting the correction for one-half the variance of log returns. (Because the regressions include a constant, the mean of the

Table 5 The real and nominal expectations hypotheses and the Cochrane-Piazzesi factor^a

A: 1952.9–2009.12			B: 1999.4–2009.12					
	$xr_{n,t+1}^{\$}$	$xr_{n,t+1}^{\$}$	$xr_{n,t+1}^{\$}$	$xr_{n,t+1}^{TIPS}$	$r_{n,t+1}^b$	$xr_{n,t+1}^{\$}$	$xr_{n,t+1}^{TIPS}$	$r_{n,t+1}^b$
$y_{n,t}^{\$} - y_{1,t}^{\$}$		2.74* (1.23)				1.88 (1.69)		
$y_{n,t}^{TIPS} - y_{1,t}^{TIPS}$							5.06* (1.54)	−4.33** (1.20)
$b_{n,t} - b_{1,t}$								7.65* (3.09)
CP_t	0.41** (0.13)	0.23 (0.13)	0.31 (0.37)	0.01 (0.37)	0.29 (0.26)	0.20 (0.38)	−0.29 (0.29)	0.57* (0.22)
p -value	0.00	0.00	0.41	0.97	0.27	0.27	0.00	0.00
R^2	0.04	0.06	0.02	0.00	0.02	0.04	0.14	0.27

^a $CP_t = -2.14 f_{1,t}^{\$} + 0.81 f_{2,t}^{\$} + 3.00 f_{3,t}^{\$} + 0.80 f_{4,t}^{\$} - 2.08 f_{5,t}^{\$}$.
Overlapping quarterly returns. Newey-West standard errors with three lags in parentheses.
* and ** denote significance at the 5% and 1% levels, respectively.
 p -value of the F-test for no predictability.

$$\begin{aligned}xr_{n,t+1}^{\$} &= r_{n,t+1}^{\$} - y_{1,t}^{\$} \\xr_{n,t+1}^{TIPS} &= r_{n,t+1}^{TIPS} - y_{1,t}^{TIPS} \\r_{n,t+1}^b &= xr_{n,t+1}^{\$} - xr_{n,t+1}^{TIPS}\end{aligned}$$

Table 6 Moments of fitted bond risk premia^a

A: Fitted risk premia U.S.				
x	$E(\hat{x})$	$\sigma(\hat{x})$	$\sigma^2(\hat{x})/\sigma^2(xr_{n,t}^{\$})$	$\sigma^2(\hat{x})/\sigma^2(xr_{n,t}^{TIPS})$
Nominal bonds	3.26	1.56	3%	
TIPS	4.16	2.70	10%	12%
Breakeven inflation	−0.91	3.24	14%	

B: Fitted risk premia UK				
x	$E(\hat{x})$	$\sigma(\hat{x})$	$\sigma^2(\hat{x})/\sigma^2(xr_{n,t}^{\$})$	$\sigma^2(\hat{x})/\sigma^2(xr_{n,t}^{TIPS})$
Nominal gilts	3.47	3.13	5%	
Inflation-linked gilts	1.66	1.84	2%	4%
Breakeven inflation	1.81	2.55	3%	

^aAnnualized (%). Columns 2 and 3 show mean and standard deviation of fitted values.
Risk premia in panel A correspond to fitted values in Table 4, panel A, columns 2, 3, and 5.
Risk premia in panel B correspond to fitted values in Table 4, panel B, columns 2, 3, and 5.
See Table 2 for definition of $xr_{n,t+1}^{\$}$ and $xr_{n,t+1}^{TIPS}$.
Abbreviation: TIPS, Treasury inflation-protected securities.

fitted values coincide with the mean excess log return reported in Table 2.) Panel A reports results for the United States, while panel B reports results for the United Kingdom.

Panel A in Table 6 shows that TIPS have had a high average risk premium over our sample period. This premium is also larger than the average risk premium on nominal

bonds, which results in a negative average breakeven inflation risk premium. Pflueger & Viceira (2011) show evidence that this negative premium is mostly driven by a positive liquidity risk premium in TIPS, not by a negative inflation risk premium in nominal Treasury bonds. In fact, panel B shows that the average breakeven inflation risk premium in the UK is positive, consistent with the notion that the inflation risk premium is positive on average. Columns 3 and 4 in each panel show that bond risk premia exhibit significant variability over time, although this variability is small relative to the overall variability of realized bond excess returns.

Figure 2 (see color insert) illustrates the time series of the fitted bond risk premia and their difference—the breakeven inflation risk premium—in the United States (panel A) and in the United Kingdom (panel B). Panel A shows that the nominal and TIPS risk premia have generally moved together, and that they exhibit significant variability over time. Bond risk premia declined during the period following the stock market crash of the early 2000s, increased during 2002 and 2003, and declined and became even negative in the subsequent period until they increased again at the onset of the recent financial crisis.

However, the breakeven inflation risk premium also shows significant time series variation, implying that the magnitude of the changes in nominal and real bond risk premia was not identical. There were times at which they even moved in opposite directions. The breakeven inflation risk premium was markedly negative at the beginning of the sample, when TIPS were first issued and investors might not have been familiar with them, and during the recent financial crisis, when the TIPS risk premium increased dramatically.

Figure 2b shows the time series of the fitted UK risk premia. The nominal, real, and breakeven risk premia have moved together over the period 1985 to 2009 and, consistent with the results shown in **Table 6**, the nominal bond risk premium has been above the real bond risk premium for most of the sample. In contrast to the U.S. bond market, both the nominal bond risk premium and the real bond risk premium shot up during the financial crisis. The nominal bond risk premium increased more than the real bond risk premium, causing the breakeven inflation risk premium to increase during the crisis. As in the case of the U.S. bond market, the risk premium on UK bonds, both real and nominal, is not necessarily positive at all times. There are periods of negative bond risk premia, most notably the turn of the 1990s for both real and nominal bonds and the period 2004–2008 for real bonds.

7. CONCLUSION

This review explores the EH of the term structure of interest rates in the U.S. and in the UK government real and nominal bond markets. It documents predictability of excess returns in inflation-indexed bonds and in nominal bonds in both markets, thus rejecting the EH, both real and nominal. Although return predictability in U.S. Treasury nominal bonds has been well-documented in the past, to our knowledge this is the first review to provide direct empirical evidence for predictability of returns in real bonds. We also find robust evidence that breakeven inflation returns, or the spread between nominal bond excess returns and inflation-indexed bond excess returns, are predictable.

The rejection of the real EH implies that investors in the inflation-indexed bond market face a time-varying risk premium that reflects a time-varying real interest rate risk

premium and possibly also a time-varying liquidity premium. The rejection of the nominal EH and particularly the rejection that expected breakeven inflation returns are constant suggest that inflation risk, and the premium that investors demand for bearing it, also varies over time. Real and nominal bond risk premia appear to be positively related to the real and nominal term spreads, respectively. When the real term spread increases, expected returns on inflation-indexed bond returns increase and, interestingly, real bonds are also expected to outperform nominal bonds. When the nominal term spread increases, expected excess returns on nominal bonds increase.

The evidence against the real and nominal EH suggests that increases in the yields of long-term bonds, whether real or nominal, do not necessarily imply that expected future short-term interest rates will rise. The increase in yields, or the decline in bond prices, could be the result of an increase in the risk of long-term bonds and the risk premium that investors demand for holding them. Thus investors should be cautious in interpreting increasing yields in long-term bonds as a signal of future higher interest rates.

In recent work, Campbell et al. (2010) show that bond risk premia can move over time and take either sign depending on whether investors see bonds as safe assets or risky assets. Our estimates show significant variation over time of real and nominal bond risk premia, with periods of positive bond risk premia and periods of negative bond risk premia, suggesting a changing investor perception of the risk of real and nominal bonds.

In particular, the risk premium on TIPS has been large on average since they were first issued in 1997, more so than the average risk premium on nominal Treasury bonds, but there have been periods where it has been negative, notably the period between 2004 and the onset of the recent financial crisis. The historical large positive average of the TIPS risk premium appears to be driven by two particular periods: the years following the creation of TIPS in the late 1990s and most recently during the recent financial crisis. Campbell et al. (2009) and Pflueger & Viceira (2011) show evidence that these episodes are linked to periods in which the TIPS market was particularly illiquid, and investors might have demanded a large liquidity premium for holding them.

Our estimates also suggest that investors demand a risk premium on nominal bonds that also varies over time. Consistent with the evidence in Campbell et al. (2010), this premium might reflect the changing perception of inflation risk by investors.

Our results suggest several directions for future research. First, they suggest a more detailed analysis of the economic sources of risk in real and nominal bonds, along the lines of Campbell et al. (2010). Second, one could also explore whether the return predictability in the inflation-indexed bond market is the result of price pressure and supply imbalances caused by limited arbitrage and preferred-habitat investors in the bond market, along the lines of the preferred-habitat hypothesis of Modigliani & Sutch (1966), formalized in Vayanos & Vila (2009) and Greenwood & Vayanos (2008, 2009).

Arguably the inflation-indexed bond market is a natural candidate to look for segmentation effects in the bond market. Just as investors might differ in their preference for bond maturities, they might also differ in their preference for holding inflation-indexed or nominal bonds. For example, some investors such as traditional defined-benefit pension funds in the United States with a mature liability structure have liabilities that are mostly nominal, whereas other investors such as less mature defined-benefit pension funds or individuals investing for retirement face liabilities that are mostly indexed. Pflueger & Viceira (2011) further explore this hypothesis.

Finally, it would be interesting to explore the implications of our findings for portfolio management and pension investing and how these implications vary by investment horizon and the investor's share of real and nominal liabilities.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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